

SCORE / LEGEND

XenaKube_Solve Study I (2026)

for smart Rubik's cube, virtual cello (SWAM physical model), and self-generating score.

SCRAMBLE—SOLVE FORM

The submitted study follows a scramble → solve arc, which acts to guide the improvisation of the performance.

CELLO—BODY

The cello is a SWAM physical model.

A turn produces a *phrase*, not a single note: pluck cluster, bowed run, sustained line, glissando, tremolo.

Phrase length, density, and register come from the active K-vertex's parameter triple. Between turns, gyro values (tilt · spin · deviation · scramble) modulate bow pressure, position, and vibrato continuously.

SOUND—COMPLEXES (C1 – C8)

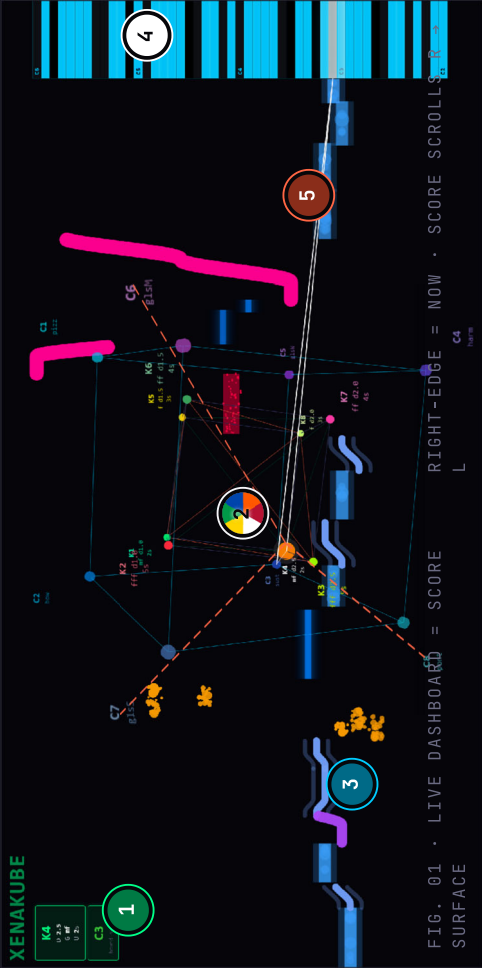
C1	Pizzicato cloud	C2	Bowed asc / desc cloud
C3	Flat bowed cloud	C4	Ionized atom — harmonics + interference
C5	Ataxic glissando	C6	Ordered glissando
C7	Flat sustained sliding	C8	Near-bridge tremolo atom

K—VERTICES · PATH V1

K1	d1.0 · gmf · u2s	K2	d1.0 · gff · u5s
K3	d2.5 · gff · u5s	K4	d2.5 · gmf · u2s
K5	d1.5 · gf · u3s	K6	d1.5 · gff · u4s
K7	d2.0 · gf · u4s	K8	d2.0 · gff · u3s

d density (events/sec) · g intensity · u duration

COMPOSER / PERFORMER HAN GEUL LEE
DURATION [VERIFY]



1 K · C CARDS

Active K-vertex (parameter triple) and C-complex (C1–C8).

2

LIVE CUBE VIEW

3D played state with translucent ghost showing the nearest of 24 S4 rotation symmetries. See 'K-vertices'.

3

PITCH SIEVE

49 cells, C2–C6. Lit briefly as pitches sound — an irregular pitch lattice that mutates as the cube moves.

4 LINE OVERLAY

White geometry linking the active cube vertex and ghost vertex to their sieve cell at the right edge where the materials are produced from.

5

C—COMPLEX LABELS

C1–C8 floating around the cube — the eight Xenakis sound-types currently assigned to vertices. See 'Sound-Complexes'.

Technical notes

GAN t4 smart cube → Web Bluetooth (Chrome) → Node relay → TypeScript engine (S₄, K_i/C_j, sieve, recognition, expression) → OSC → Max/MSP hosting SWAM Cello 3 → MIDI echo back to browser → live score.

CUBE > RELAY > ENGINE > SWAM > SCORE